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Basic Reproduction Number of a Multi-Patch SVI Model Represented as a Star Graph Topology

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Abstract. A mathematical model of Susceptible-Vaccinated-Infected (SVI) disease multi-patch model will be introduced in this article. Interaction between cities which are represented as a star graph involved in the model along with vaccination and medical treatment intervention. Human mobility is assumed to be constant between satellite patch and center patch. Basic reproduction number as the endemic indicator is constructed using a graph-theoretical approach. We find that basic reproduction number of the model is given by the reciprocal of the smallest positive root of a two degree polynomial. This method can be an alternative to determine a form of basic reproduction number for a complex system of ODE model.

Keywords: Susceptible-Vaccinated-Infected, patch, human mobility, basic reproduction number

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INTRODUCTION

An SIR (Susceptible-Infected-Recovered) model is a very first simple model that can be used to describe the transmission of disease which can be transmitted directly among human population. The model is constructed involving new born, natural death rate, natural recovery from disease and transmission process. A mathematical analysis of the system about the existence of equilibrium points and their local stability become a standard routine when analyzing this model. The interaction between S, I and R subpopulation can be seen as a transmission diagram given by Fig. 1.

One other important numbers which very crucial to be understood is the basic reproduction number, which denoted by $\mathcal{R}_0$. In many mathematical model, the existence and local stability of the equilibrium points depend on the value of $\mathcal{R}_0$. There are many method to calculate basic reproduction number, such as with next-generation matrix [1] and with graph theory [2].

Many mathematical models have been introduced based on this SIR model in many kind of disease, such as dengue [3–5], respiratory-type disease [6–8], Tuberculosis [9–11] , and some others. In this article, a mathematical model of extension of the SIR model into SVI (Susceptible-Vaccinated-Infected) model will be introduced involving vaccination and medical treatment intervention. Human mobility also included into the model as a multi-patch model which will be represented as a connected graph.

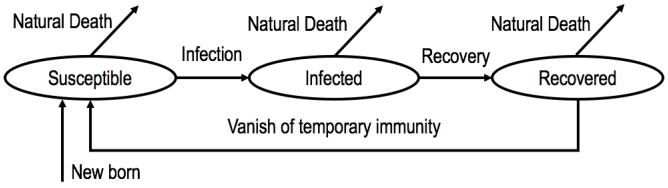


FIGURE 1. Interaction diagram for simple SIR model

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In the next section, a construction of the SVI model involving human mobility which is represented as a connected graph between cities will be given. Construction of the basic reproduction number for simple two patches and then followed by n-patches model will be given later.

MATHEMATICAL MODEL

The first mathematical model that could be used to describe a spread of disease with SI model was introduced by Kermack and McKendrick in 1927. This model is known as The Susceptible-Infected-Recovered (SIR) model. From SIR model, we construct a mathematical model to interpret the spread of disease that might transmitted directly between humans with 3-dimension autonomous ordinary differential equations which is represented by

$$\begin{aligned} \frac{dS}{dt} &= A - \beta S I - (u_1 + \mu) S + \delta V + (\gamma_0 + \gamma_1 u_2) I, \\ \frac{dV}{dt} &= u_1 S - (\delta + \mu) V \\ \frac{dI}{dt} &= \beta S I - (\gamma_0 + \gamma_1 u_2 + \mu) I, \end{aligned} \tag{1}$$

where t presented as time variable. This model describes the interactions among the humans. The population is divided into three classes: Susceptible humans, Vaccinated humans, and Infected humans, with size S, V, and I respectively. The parameter A is the rate at which new humans are born and die at the rate μ. In this model, some population of susceptible class will get a vaccination (u_1) which will give them a temporary immunity ($1/\delta$) from infection of disease. Therefore, only susceptible human who do not get a vaccination will have a probability success to get a disease infection which described by the term of $\beta S I$, where β represented as per-capita of success probability of infection. People might recovered from disease depends on their natural immune system and will get recovered much faster if they get medical treatment. If some portion of infected people will get a medical treatment, let say with a constant portion of u_2, then their natural recovery rate will be improved from γ_0 into $\gamma_0 + \gamma_1$. In the other hand, people who do not get a medical treatment with portion of $1 - u_2$ will get recovered only depend on their natural immunity γ_0.

There is other significant factor that also play an important role in determining the spread of disease, that is spatial factor and human mobility. The human mobility is also important to transmit dynamics spatially [12]. The human mobility can be described as a directed graph. In this paper, we use the star graph to illustrate how humans can do mobility from one patch to other.

Consider an SVI model with n patches which susceptible (S), vaccinated (V), and infected (I) in each patches can travel going through a central patch. The mobility of human in this model could be illustrated as a star graph with a center patch and $n - 1$ patches around it as a satellite patches is shown in Fig. 2.

Humans can travel from patch 1 to patch j and from patch j to patch 1 at the rate $m_{j,1}$ and $m_{1,j}$, respectively. Unlike human in satellite patch which only can do mobility into center patch, humans in center patch can travel to

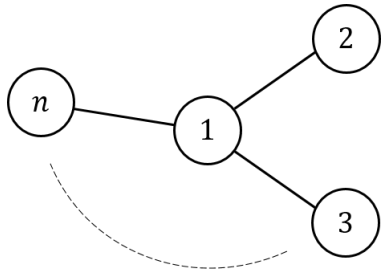


FIGURE 2. Star graph with n patches

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every satellite patch j. The dynamics at the center (node 1) is given by

$$\begin{aligned}
\frac{dS_1}{dt} &= A - \beta S_1 I_1 - (u_1 + \mu) S_1 + \delta V_1 + (\gamma_0 + \gamma_1 u_2) I_1 - \sum_{j=2}^{n} m_{j,1} S_1 + \sum_{j=2}^{n} m_{1,j} S_j, \\
\frac{dV_1}{dt} &= u_1 S_1 - (\delta + \mu) V_j - \sum_{j=2}^{n} m_{j,1} V_1 + \sum_{j=2}^{n} m_{1,j} V_j, \\
\frac{dI_1}{dt} &= \beta S_1 I_1 - (\gamma_0 + \gamma_1 u_2 + \mu) I_1 - \sum_{j=2}^{n} m_{j,1} I_1 + \sum_{j=2}^{n} m_{1,j} I_j,
\end{aligned} \tag{2}$$

while for the satellite patches $j = 2, 3, \cdots, n$ is given by

$$\begin{aligned}
\frac{dS_j}{dt} &= A - \beta S_j I_j - (u_1 + \mu) S_j + \delta V_j + (\gamma_0 + \gamma_1 u_2) I_j - m_{1,j} S_j + m_{j,1} S_1, \\
\frac{dV_j}{dt} &= u_1 S_j - (\delta + \mu) V_j - m_{1,j} V_j + m_{j,1} V_1, \\
\frac{dI_j}{dt} &= \beta S_j I_j - (\gamma_0 + \gamma_1 u_2 + \mu) I_j - m_{1,j} I_j + m_{j,1} I_1.
\end{aligned} \tag{3}$$

BASIC REPRODUCTION NUMBER ($\mathcal{R}_0$)

The basic reproduction number ($\mathcal{R}_0$) is the most important quantity in infectious disease epidemiology which can described the behavior of the dynamical system respect to each parameters that involved in the model. $\mathcal{R}_0$ express an expected number of secondary infection cases caused by initial infection in a completely susceptible population during one infection period [1]. Based on the definition of the $\mathcal{R}_0$, whenever this value is larger than one, then the disease will spreads among population and push the system into an endemic situation. In the other hand, when this $\mathcal{R}_0$ is smaller than one, then the disease will disappear [13]. Calculating $\mathcal{R}_0$ using the next-generation matrix sometimes difficult whenever the dimension of the model increasing. One option to solve this problem is to using the graph-theoritic method to calculate the $\mathcal{R}_0$ in a simpler way [2].

Based on de-Camino-Beck *et al.*'s research in 2009 [2], the algorithm to compute $\mathcal{R}_0$ with graph-theoretic method are:

1. As a first step, let the infected subsystem of a complete disease model linearized in the disease free equilibrium (DFE) and let call this matrix as Jacobian matrix $\mathbf{J}$.
2. Let construct the transmission ($\mathbf{F}$) and transition ($\mathbf{V}$) part of a complete model, by decomposing $\mathbf{F}$ matrix where $\mathbf{F} = \mathbf{F} - \mathbf{V}$.
3. Consider matrix $\mathbf{F}\lambda^{-1} - \mathbf{V}$ then build a directed graph from this matrix. Please note that the λ is the root of $\mathbf{F}\lambda^{-1} - \mathbf{V} = 0$, where this λ will become the basic reproduction number.
4. Do the reduction process on the directed graph that formed from matrix $\mathbf{F}\lambda^{-1} - \mathbf{V}$. There are two rules on this reduction process:
   (a) Rule 1: Creating a trivial node.
   To reduce the loop $-a_{ii} < 0$ to -1 at node i, every arc entering node-i has weight divided by a_{ii} as shown in Fig. 3. Please note that a_{ij} is the edge that connect node-i and node-j.
   (b) Rule 2: Elimination of arcs through a trivial node.
   For a trivial node i on a path $j \to i \to k$, the two arcs are replaced by $j \to k$ with weight equal to the product of weights on arc $j \to i$ and $i \to k$ as shown in Fig. 4.

In the next section, we will construct the basic reproduction number for two patched model of system (Equation 2 and 3) with only involving a center patch and one satellite patch.

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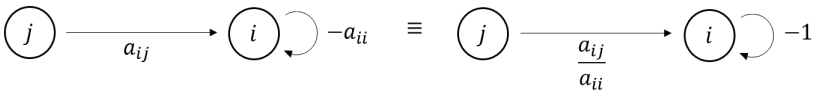


FIGURE 3. Creating a trivial node

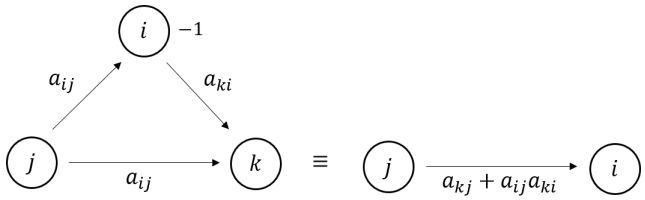


FIGURE 4. Elimination of arcs through a trivial node

BASIC REPRODUCTION NUMBER ($\mathcal{R}_0$) FOR TWO PATCH SYSTEM

Consider the SVI model with mobility in a two patches system where center patch is given by

$$\begin{aligned}
\frac{dS_1}{dt} &= A - \beta S_1 I_1 - (u_1 + \mu) S_1 + \delta V_1 + (\gamma_0 + \gamma_1 u_2) I_1 - m_{2,1} S_1 + m_{1,2} S_2, \\
\frac{dV_1}{dt} &= u_1 S_1 - (\delta + \mu) V_1 - m_{2,1} V_1 + m_{1,2} V_2, \\
\frac{dI_1}{dt} &= \beta S_1 I_1 - (\gamma_0 + \gamma_1 u_2 + \mu) I_1 - m_{2,1} I_1 + m_{1,2} I_2.
\end{aligned} \tag{4}$$

while for the satellite patch is given by

$$\begin{aligned}
\frac{dS_2}{dt} &= A - \beta S_2 I_2 - (u_1 + \mu) S_2 + \delta V_2 + (\gamma_0 + \gamma_1 u_2) I_2 - m_{1,2} S_2 + m_{2,1} S_1, \\
\frac{dV_2}{dt} &= u_1 S_2 - (\delta + \mu) V_2 - m_{1,2} V_2 + m_{2,1} V_1, \\
\frac{dI_2}{dt} &= \beta S_2 I_2 - (\gamma_0 + \gamma_1 u_2 + \mu) I_2 - m_{1,2} I_2 + m_{2,1} I_1.
\end{aligned} \tag{5}$$

From Equation 4 and 5 we can linearize the infected subsystem :

$$\mathbf{J} = \begin{pmatrix} \beta \bar{S}_1 - P_1 & m_{1,2} \\ m_{2,1} & \beta \bar{S}_2 - P_2 \end{pmatrix} \tag{6}$$

where $P_1 = (\gamma_0 + \gamma_1 u_2 + \mu + m_{2,1} + m_{3,1} + \cdots + m_{k,1})$ and $P_2 = (\gamma_0 + \gamma_1 u_2 + \mu + m_{1,2})$. From Equation 6 we can decompose matrix **J** into **F** - **V** where **F** and **V** given by

$$\mathbf{F} = \begin{pmatrix} \beta \bar{S}_1 & 0 \\ 0 & \beta \bar{S}_2 \end{pmatrix}, \mathbf{V} = \begin{pmatrix} (\gamma_0 + \gamma_1 u_2 + \mu + m_{2,1}) & -m_{1,2} \\ -m_{2,1} & (\gamma_0 + \gamma_1 u_2 + \mu + m_{1,2}) \end{pmatrix}. \tag{7}$$

From Equation 7, consider $\mathbf{F}\lambda^{-1} - \mathbf{V}$ matrix

$$\begin{aligned}
\mathbf{F}\lambda^{-1} - \mathbf{V} &= \begin{pmatrix} \beta \bar{S}_1 & 0 \\ 0 & \beta \bar{S}_2 \end{pmatrix} \lambda^{-1} - \begin{pmatrix} (\gamma_0 + \gamma_1 u_2 + \mu + m_{2,1}) & -m_{1,2} \\ -m_{2,1} & (\gamma_0 + \gamma_1 u_2 + \mu + m_{1,2}) \end{pmatrix} \\
&= \begin{pmatrix} \beta \bar{S}_1 \lambda^{-1} - (\gamma_0 + \gamma_1 u_2 + \mu + m_{2,1}) & m_{1,2} \\ m_{2,1} & \beta \bar{S}_2 \lambda^{-1} - (\gamma_0 + \gamma_1 u_2 + \mu + m_{1,2}) \end{pmatrix}
\end{aligned} \tag{8}$$

and build a directed graph as shown in Fig. 5. Using Rule 1 and 2 in the Fig. 5, this graph reduced into Fig. 6.

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The weight of patch I_1 in Fig. 6 is described as $\det\left(\mathbf{F}\lambda^{-1} - \mathbf{V}\right)$ and be written as polynomial equation of $\det\left(\mathbf{F}\lambda^{-1} - \mathbf{V}\right) = 0$.

$$\beta\bar{S_1}\lambda^{-1} - (\gamma_0 + \gamma_1 u_2 + \mu + m_{2,1}) + \frac{m_{1,2}m_{2,1}}{(\gamma_0 + \gamma_1 u_2 + \mu + m_{1,2}) - \beta\bar{S_2}\lambda^{-1}} = 0. \tag{9}$$

Therefore, the $\mathcal{R}_0$ is taken from the smallest root positive root of Equation 9 which read as :

$$\mathcal{R}_0 = 1/4\,\frac{\left(P_1\beta_2 + P_2\beta_1 + \sqrt{{P_2}^2{\beta_1}^2 - 2\beta_2\left(P_1P_2 - 2\,m_{1,2}m_{2,1}\right)\beta_1 + {P_1}^2{\beta_2}^2}\right)^2 A^2}{\left(P_1P_2 - m_{1,2}m_{2,1}\right)^2\mu^2} \tag{10}$$

BASIC REPRODUCTION NUMBER ($\mathcal{R}_0$) FOR n PATCHES

Consider the SVI model with mobility represented as star graph with n patches given by system 2 and 3. Linearize the infected subsystem gave us

$$\mathbf{J} = \begin{pmatrix} \beta\bar{S}_1 - Q_1 & m_{1,2} & m_{1,3} & \cdots & m_{1,n} \\ m_{2,1} & \beta\bar{S}_2 - Q_2 & 0 & \cdots & 0 \\ m_{3,1} & 0 & \beta\bar{S}_3 - Q_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ m_{n,1} & 0 & 0 & \cdots & \beta\bar{S}_n - Q_n \end{pmatrix} \tag{11}$$

where

$$\begin{aligned} Q_1 &= (\gamma_0 + \gamma_1 u_2 + \mu + m_{2,1} + m_{3,1} + \cdots + m_{n,1}) \\ Q_2 &= (\gamma_0 + \gamma_1 u_2 + \mu + m_{1,2}) \\ Q_3 &= (\gamma_0 + \gamma_1 u_2 + \mu + m_{1,3}) \\ Q_n &= (\gamma_0 + \gamma_1 u_2 + \mu + m_{1,n})\,. \end{aligned} \tag{12}$$

From Equation 11 we can decompose matrix **J** into **F** - **V** where **F** and **V** given by

$$\mathbf{F} = \begin{pmatrix} \beta\bar{S}_1 & 0 & 0 & \cdots & 0 \\ 0 & \beta\bar{S}_2 & 0 & \cdots & 0 \\ 0 & 0 & \beta\bar{S}_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \beta\bar{S}_n \end{pmatrix}, \mathbf{V} = \begin{pmatrix} Q_1 & -m_{1,2} & -m_{1,3} & \cdots & -m_{1,n} \\ -m_{2,1} & Q_2 & 0 & \cdots & 0 \\ -m_{3,1} & 0 & Q_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -m_{n,1} & 0 & 0 & \cdots & Q_n \end{pmatrix} \tag{13}$$

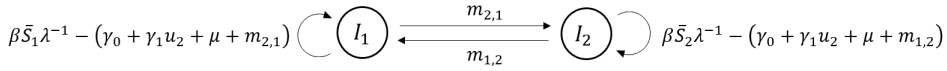


FIGURE 5. Directed star graph with two patches

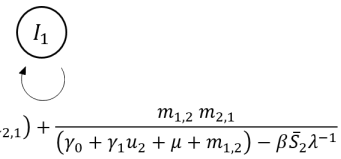


FIGURE 6. Reduction patch I_2

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From Equation 13, consider $\mathbf{F}\lambda^{-1} - \mathbf{V}$ matrix

$$\begin{aligned}\mathbf{F}\lambda^{-1} - \mathbf{V} &= \begin{pmatrix} \beta\bar{S}_1 & 0 & 0 & \cdots & 0 \\ 0 & \beta\bar{S}_2 & 0 & \cdots & 0 \\ 0 & 0 & \beta\bar{S}_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \beta\bar{S}_n \end{pmatrix}\lambda^{-1} - \begin{pmatrix} Q_1 & -m_{1,2} & -m_{1,3} & \cdots & -m_{1,n} \\ -m_{2,1} & Q_2 & 0 & \cdots & 0 \\ -m_{3,1} & 0 & Q_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -m_{n,1} & 0 & 0 & \cdots & Q_n \end{pmatrix} \\ &= \begin{pmatrix} \beta\bar{S}_1\lambda^{-1} - Q_1 & m_{1,2} & m_{1,3} & \cdots & m_{1,n} \\ m_{2,1} & \beta\bar{S}_2\lambda^{-1} - Q_2 & 0 & \cdots & 0 \\ m_{3,1} & 0 & \beta\bar{S}_3\lambda^{-1} - Q_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ m_{n,1} & 0 & 0 & \cdots & \beta\bar{S}_n\lambda^{-1} - Q_n \end{pmatrix}, \end{aligned} \tag{14}$$

and build a directed graph as shown in Fig. 7.

Using role 1 and 2 in second section, Fig. 7 can be reduced into Fig. 8. The weight of patch I_1 in Fig. 8 is described as $\det\left(\mathbf{F}\lambda^{-1} - \mathbf{V}\right)$ and can be written as polynomial equation of $\det\left(\mathbf{F}\lambda^{-1} - \mathbf{V}\right) = 0$ which given by

$$\begin{aligned}\beta\bar{S}_1\lambda^{-1} - (\gamma_0 + \gamma_1 u_2 + \mu + m_{2,1} + m_{3,1} + \cdots + m_{n,1}) &+ \frac{m_{1,2}m_{2,1}}{(\gamma_0 + \gamma_1 u_2 + \mu + m_{1,2}) - \beta\bar{S}_2\lambda^{-1}} \\ &+ \frac{m_{1,3}m_{3,1}}{(\gamma_0 + \gamma_1 u_2 + \mu + m_{1,3}) - \beta\bar{S}_3\lambda^{-1}} \\ &+ \cdots \\ &+ \frac{m_{1,n}m_{n,1}}{(\gamma_0 + \gamma_1 u_2 + \mu + m_{1,n}) - \beta\bar{S}_n\lambda^{-1}} = 0. \end{aligned} \tag{15}$$

To determine the $\mathcal{R}_0$, Equation 15 should be solved respect to λ. The smallest root of this polynomial will become the $\mathcal{R}_0$.

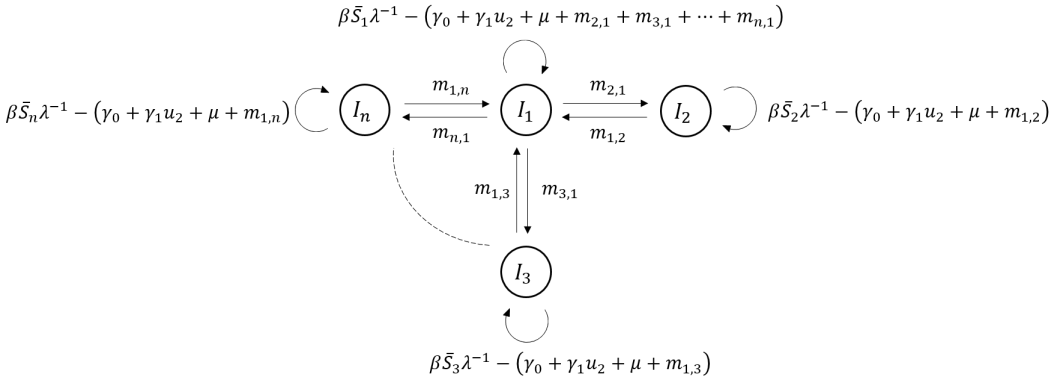


FIGURE 7. Directed star graph with n patches

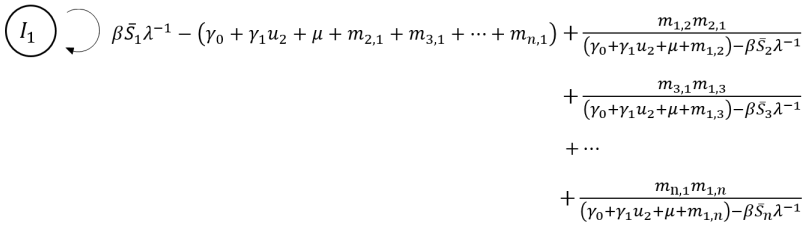


FIGURE 8. Reduction patch I_2, I_3,$\cdots$,I_n

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CONCLUSION

A mathematical model of infectious disease that might transmitted directly between humans with vaccination and medical treatment have been introduced in this article. Interaction between city and villages also involved in the model which constructed as a multi-patch model, where the interaction between patches is represented as a star graph. The basic reproduction number as the endemic indicator have been founded as a reciprocal of the smallest root of a two degree polynomial. Unlike with the next-generation matrix approach which will be very difficult to calculate basic reproduction number for a multi patch system when number of patches is large, with graph theoretical approach, no mater how many patches are involved in the model, the basic reproduction number can always be found only from two degree polynomial only.

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